

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative modeler with industry experience in financial-risk model validation, insurance pricing, and scalable statistical model development. Combines stochastic processes, likelihood-based inference, econometrics, simulation, and statistical diagnostics with reproducible Python/PyTorch research systems. Translates model assumptions, performance, uncertainty, and limitations into clear findings for technical and business stakeholders.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Feb 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), R, SQL, C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, cached artifacts, and restartable analytical pipelines

Quantitative Methods: Stochastic processes, probability, MLE/GLS, regression, GLM/GBM, econometrics, fixed effects, propensity-score matching, Monte Carlo simulation, hypothesis testing, numerical optimization, model selection and validation, sensitivity analysis, feature attribution, and statistical diagnostics

QUANTITATIVE INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring requirement into explicit hypotheses, likelihood-ratio boundaries, and controlled stopping decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate Type I/II error, boundary-crossing probabilities, stopping behavior, uncertainty, and model limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ property-risk records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Evaluated performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

RESEARCH AND QUANTITATIVE MODELING

Ph.D. Dissertation Research - Scalable Statistical Model Development and Diagnostics

Sep 2024 - Present

- **Scalable Model Development:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered local conditioning to fit up to 145,008 observations per day.
- **Assumption Review and Diagnostics:** Compared alternative covariance specifications and designed scale- and frequency-resolved tools to determine whether discrepancies arise from low-frequency structure, high-frequency behavior, noise, missingness, or model misspecification.
- **Research Engineering:** Implemented reusable Python/PyTorch workflows for data-quality filtering, estimation, simulation, numerical optimization, cached outputs, and restartable CPU/GPU and HPC execution; integrated compiled C++ ordering routines through pybind11.

Econometric Study - Earned Income Tax Credit and Household Labor Supply

2021

- Estimated the association between EITC exposure and household labor supply using longitudinal observational data, fixed-effects regression, and propensity-score matching with household income and demographic covariates.

ADDITIONAL

Coursework: Probability Theory, Stochastic Processes, Machine Learning, Linear Algebra, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)